

graph is a negative sine graph.)

- 15** Five particles have speeds of $2u$, $5u$, $10u$, $11u$ and $15u$. Which of the following statements is correct?
- A** The root mean square speed is equal to the mean speed.
 - B** The root mean square speed exceeds the mean speed by more than $1u$.
 - C** The root mean square speed exceeds the mean speed by less than $1u$.
 - D** The mean speed exceeds the root mean square speed by more than $1u$.

Ans: B

$$\text{r.m.s. speed} = \sqrt{((2u)^2 + (5u)^2 + (10u)^2 + (11u)^2 + (15u)^2 / 5)} = 9.75 u$$

$$\text{mean speed} = (2u + 5u + 10u + 11u + 15u) / 5 = 8.6 u$$

- 16** Latent heat of vaporisation is the energy required to